

Product Requirements Document

Airport Baggage Handling System

Document status: Draft

Version: 0.1

Product: Airport Baggage Handling System

Primary architecture model: SAIL BHS model

Prepared for: Product, architecture, engineering, airport operations, and baggage operations stakeholders

1. Product Summary

The Airport Baggage Handling System is responsible for coordinating the software-controlled movement, identification, routing, tracking, and operational management of baggage within an airport baggage handling environment.

The system supports outbound baggage handling from airline source-message ingestion through physical induction, identity correlation, route determination, sortation, equipment coordination, operational monitoring, diagnostics, and exception handling.

The system integrates with airline systems, airport operational systems, hold baggage screening systems, PLCs, sensors, and human operators.

2. Problem Statement

Airport baggage handling requires coordination across physical equipment, airline data, airport schedules, security screening, operational rules, and human intervention.

Without a coherent system boundary and operating model, baggage flow can become difficult to reason about. Bags may be inducted before source data is available, routed with incomplete context, delayed by unreadable tags, blocked by equipment faults, or lose traceability across handoffs.

The Baggage Handling System must provide a reliable software foundation for processing outbound bags, coordinating with external systems, and giving operators enough visibility to detect and resolve problems.

3. Goals

Goal	Description
Process outbound baggage reliably	Support the full outbound baggage flow from source-message receipt through routing and sortation.
Maintain baggage traceability	Capture events that allow operators to understand where a bag is, where it has been, and what decisions were made.
Coordinate physical movement	Exchange commands and status with PLCs, sensors, and equipment-control systems.
Support operational intervention	Allow operators to detect, diagnose, and resolve baggage exceptions.
Integrate with airport and airline systems	Use flight, passenger, baggage, schedule, and security context to support routing decisions.
Support progressive architectural decomposition	Allow the system to be modeled and implemented through subsystems, service units, interfaces, interactions, and process diagrams.

4. Non-Goals

Non-goal	Description
Airline reservation management	The BHS does not manage reservations, ticketing, or passenger booking records.
Airline DCS ownership	The BHS consumes data from the Departure Control System but does not replace it.
Security screening algorithm ownership	The BHS coordinates with Hold Baggage Screening but does not implement explosive-detection logic.
Mechanical conveyor design	The BHS integrates with physical equipment but does not define mechanical layout or conveyor engineering.
Full airport operations platform	The BHS uses operational data but does not replace the AODB or airport resource-management systems.

Non-goal	Description
Aircraft loading optimization	Loading and reconciliation may be integrated, but aircraft load planning is not the core responsibility unless explicitly added to scope.

5. Users and Boundary Participants

Participant	Type	Role
Check-in Agent	Human user	Initiates or supports baggage acceptance and resolves intake issues.
Self-Service Kiosk / Bag Drop	External system	Initiates passenger-driven baggage acceptance.
Baggage Handler	Human user	Handles bags physically and may interact with exception or status workflows.
System Operator	Human user	Monitors system health, routing state, equipment status, and exceptions.
Airline Departure Control System	External system	Provides baggage source messages and passenger/flight/bag data.
Airport Operational Database	External system	Provides schedule, gate, stand, destination, and operational context.
Hold Baggage Screening System	External system	Screens baggage and returns screening status.
PLCs	Device / external control system	Execute physical movement commands and report equipment state.
Sensors	Sensors	Provide physical bag observations such as reads, presence, dimensions, or movement events.

6. System Scope

In Scope

Capability	Description
Receive baggage source messages	Ingest and validate source messages from airline systems.
Induct physical bags	Accept physical bags into the handling flow from supported induction points.
Identify bags	Associate physical bags with tag, flight, passenger, and routing context where available.
Route bags	Determine the next destination or handling path for each bag.
Coordinate sortation	Support movement toward the correct downstream sortation or makeup destination.
Coordinate screening	Exchange screening requests/results with HBS where required.
Track bag state	Maintain event-based traceability across source messages, induction, routing, and equipment movement.
Monitor equipment and operations	Provide visibility into operational state, equipment health, and faults.
Handle exceptions	Support manual intervention for missing data, unreadable tags, routing failures, equipment issues, and screening-related exceptions.
Publish operational events	Emit events needed by downstream systems, operators, reports, and diagnostics.

Out of Scope / TBD

Area	Status
Arrival baggage / reclaim	TBD. Current modeling focus appears to be outbound baggage.
Transfer baggage-specific optimization	TBD. May be added as a routing scenario.

Area	Status
Baggage reconciliation ownership	TBD. BHS may integrate with BRS but does not necessarily own reconciliation.
Predictive analytics / optimization	Future enhancement unless required by stakeholders.

7. Major Subsystems

The current level-0 hierarchy identifies the BHS as a system composed of major internal subsystems and a datastore/event backbone. The current deck identifies five subsystems inside the BHS and introduces the High Speed Event Data Bus as a datastore.

Subsystem	Purpose
External Integration Gateway	Handles integrations with airline and airport systems such as DCS, AODB, HBS, and other external systems.
Routing & Sortation Engine	Determines where bags should go next based on source data, bag state, routing rules, screening state, and equipment conditions.
Tracking & Diagnostics	Maintains baggage tracking, operational diagnostics, and visibility into bag movement and system behavior.
Equipment Control Node	Coordinates with PLCs, sensors, and material-handling equipment. This is expected to be decomposed further into service units.
Operations UI	Provides user-facing operational views for monitoring, intervention, exception handling, and diagnostics.
High Speed Event Data Bus	Provides event movement, buffering, and distribution across the BHS.

8. Key User Journeys / System Behaviors

8.1 Process Outbound Bag

The core outbound scenario coordinates source-message data, physical bag induction, identity, routing, and equipment movement.

High-level flow:

1. BHS receives or already has baggage source-message data.
2. A physical bag is inducted into the system.
3. The bag is identified through tag, sensor, or manual input.
4. The system correlates bag identity with flight and routing context.
5. The system determines the bag's next route.
6. The equipment-control path executes or coordinates physical movement.
7. Events are published for tracking, diagnostics, and operational visibility.
8. Exceptions are surfaced when the bag cannot proceed automatically.

8.2 Receive Baggage Source Message

The system receives a baggage source message from the airline DCS or equivalent external source. This behavior is currently modeled as a simple interaction with an associated characteristic map in the deck, which is used to capture architectural details that do not belong on a basic flow diagram.

8.3 Induct Physical Bag

The system receives a physical induction event from a belt, bag drop, PLC, sensor, or related equipment-control source.

8.4 Route Bag

The system determines and communicates the correct handling path for a bag based on available operational context.

8.5 Monitor and Resolve Exceptions

Operators observe exceptions and intervene when automated handling cannot proceed.

9. Functional Requirements

ID	Requirement
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FR-001	The system shall receive baggage source messages from airline or airport systems.
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ID Requirement

- FR-002 The system shall validate received source messages for required baggage, passenger, flight, and routing fields.
- FR-003 The system shall record accepted source-message events.
- FR-004 The system shall receive physical bag induction events from supported induction points.
- FR-005 The system shall associate physical bag observations with known baggage source data where possible.
- FR-006 The system shall support bag identification using barcode, RFID, sensor, or manual encoding input.
- FR-007 The system shall determine whether a bag has enough information to proceed automatically.
- FR-008 The system shall route bags based on flight, destination, screening status, operational rules, and equipment availability.
- FR-009 The system shall coordinate with Hold Baggage Screening where screening is required.
- FR-010 The system shall receive screening results and update bag state accordingly.
- FR-011 The system shall communicate route or movement decisions to equipment-control components.
- FR-012 The system shall receive equipment and sensor status relevant to bag movement.
- FR-013 The system shall publish bag events to the High Speed Event Data Bus.
- FR-014 The system shall maintain a traceable history of significant bag events.

ID	Requirement
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| FR-015 | The system shall expose operational status to system operators. |
| FR-016 | The system shall expose baggage exceptions to authorized operational users. |
| FR-017 | The system shall allow authorized users to resolve or override supported exception states. |
| FR-018 | The system shall support diagnostics for failed integrations, missing source messages, unreadable bag identity, route failures, and equipment faults. |
| FR-019 | The system shall prevent unauthorized users from performing operational control actions. |
| FR-020 | The system shall preserve enough event history to support post-incident investigation. |
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10. Non-Functional Requirements

Category	Requirement
Availability	The system should support continuous baggage operations during airport operating periods.
Reliability	Bag events, route decisions, and exception states must not be silently lost.
Latency	Routing and equipment-related decisions must be fast enough to support physical baggage movement without unnecessary delay.
Resilience	The system should degrade gracefully when external systems, sensors, or integrations are unavailable.
Observability	Operators and support teams must be able to diagnose integration failures, equipment issues, routing failures, and event delays.
Security	Operational controls must be limited to authorized users and roles.

Category	Requirement
Auditability	Operator actions, route decisions, source-message handling, and exception resolution should be auditable.
Traceability	The system should retain a bag-level history sufficient to reconstruct key processing decisions.
Scalability	The system should support expected airport baggage volume and peak operating periods.
Maintainability	Subsystems should have clear responsibilities and interfaces so they can be evolved independently.

11. Data and Events

Data / Event	Description
Baggage Source Message	Message containing bag, passenger, flight, and source context from airline systems.
Bag Induction Event	Event indicating that a physical bag entered the handling flow.
Bag Identity Observation	Barcode, RFID, or manual encoding observation used to identify a bag.
Bag Movement Event	Event describing movement or presence at a point in the material-handling system.
Routing Decision	Decision identifying the next destination or handling path for a bag.
Screening Request	Request or handoff indicating a bag requires screening.
Screening Result	Screening outcome such as clear, alarm, reject, or rescreen.
Equipment Status Event	Status from PLCs, sensors, conveyors, diverters, or related equipment.
Exception Event	Event indicating a problem requiring automated or manual resolution.

Data / Event	Description
Operator Action Event	Audit event recording manual intervention or operational control action.

12. Integrations

Integration	Direction	Purpose
Airline DCS	Inbound / outbound as needed	Receive baggage source messages and possibly send baggage status updates.
AODB	Inbound	Receive flight schedule and operational context.
HBS	Bidirectional	Coordinate screening requests and screening results.
PLCs / Equipment Control	Bidirectional	Send movement commands or route instructions and receive equipment status.
Sensors	Inbound	Receive bag identity, presence, movement, or measurement observations.
Operations UI	Bidirectional	Expose monitoring, diagnostics, exception handling, and operator controls.
High Speed Event Data Bus	Internal	Distribute baggage events, equipment events, and system events.

13. Acceptance Criteria

ID	Acceptance criterion
AC-001	A baggage source message can be received, validated, and recorded.
AC-002	A physical bag induction event can be correlated to source-message data when matching information is available.

ID	Acceptance criterion
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AC-003	A bag can be routed automatically when required source data, operational data, and equipment state are available.
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AC-004	A bag with missing or unreadable identity enters an exception state.
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AC-005	A screening-required bag is sent through the appropriate screening coordination flow.
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AC-006	Screening results affect the bag's subsequent routing state.
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AC-007	Equipment status can be received and made visible to operational users.
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AC-008	Operators can view active exceptions.
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AC-009	Authorized users can resolve supported exception states.
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AC-010	Significant baggage events are published to the event backbone.
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AC-011	Bag history can be reconstructed from recorded events.
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AC-012	Unauthorized users cannot perform restricted operational actions.
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14. Risks

Risk	Impact	Mitigation
Source messages arrive late or are missing	Bags may be inducted without enough routing context.	Support pending/exception states and late correlation.

Risk	Impact	Mitigation
Barcode/RFID reads fail	Bags may require manual encoding or exception handling.	Support redundant identity observations and manual encoding.
Equipment state is stale or unavailable	Routing decisions may be unsafe or ineffective.	Require equipment-state validation before routing decisions.
Event bus delays or failures	Tracking and downstream consumers may lose visibility.	Use durable event handling and monitoring.
Screening system delays	Bags may be delayed or incorrectly released.	Model screening state explicitly and require valid outcomes before routing.
Operator override misuse	Incorrect manual actions may disrupt bag flow.	Use authorization, audit, and constrained override workflows.
Overloaded diagrams or unclear decomposition	Architecture becomes difficult to maintain.	Continue using SAIL progressive disclosure and subsystem/service-unit decomposition.

15. Open Questions

Question	Why it matters
Is arrival/reclaim baggage in scope for this PRD, or is this PRD limited to outbound baggage?	Changes interactions, participants, routing rules, and success metrics.
Is BRS an external system or part of the BHS scope?	Determines ownership of reconciliation behavior.
Does the BHS command PLCs directly, or does it integrate with an intermediate equipment-control layer?	Affects subsystem boundaries and interface definitions.
What are the required latency targets for route decisions and equipment commands?	Needed for measurable NFRs.

Question**Why it matters**

What are the required event-retention periods?

Needed for audit, compliance, and datastore design.

What operational actions require role-based authorization?

Needed for security and audit design.

What degraded modes must be supported when DCS, AODB, HBS, sensors, or PLCs are unavailable?

Drives resilience requirements and process flows.

Which process interactions require full process diagrams first?

Helps prioritize the next SAIL modeling work.